PYTHON PROBLEM-SOLVING ASSIGNMENT

Flowcharts • Pseudocode • Sample Input • Sample Output

Topics covered	10 applied programming problems
Included for each problem	Aim, assumptions/formulas, flowchart, pseudocode, sample input and sample output
Currency	Indian Rupees (■), unless a foreign currency is specified

Note: All rates, thresholds and policies in this document are illustrative assumptions for learning. Replace them with the exact values supplied by your instructor if they differ.

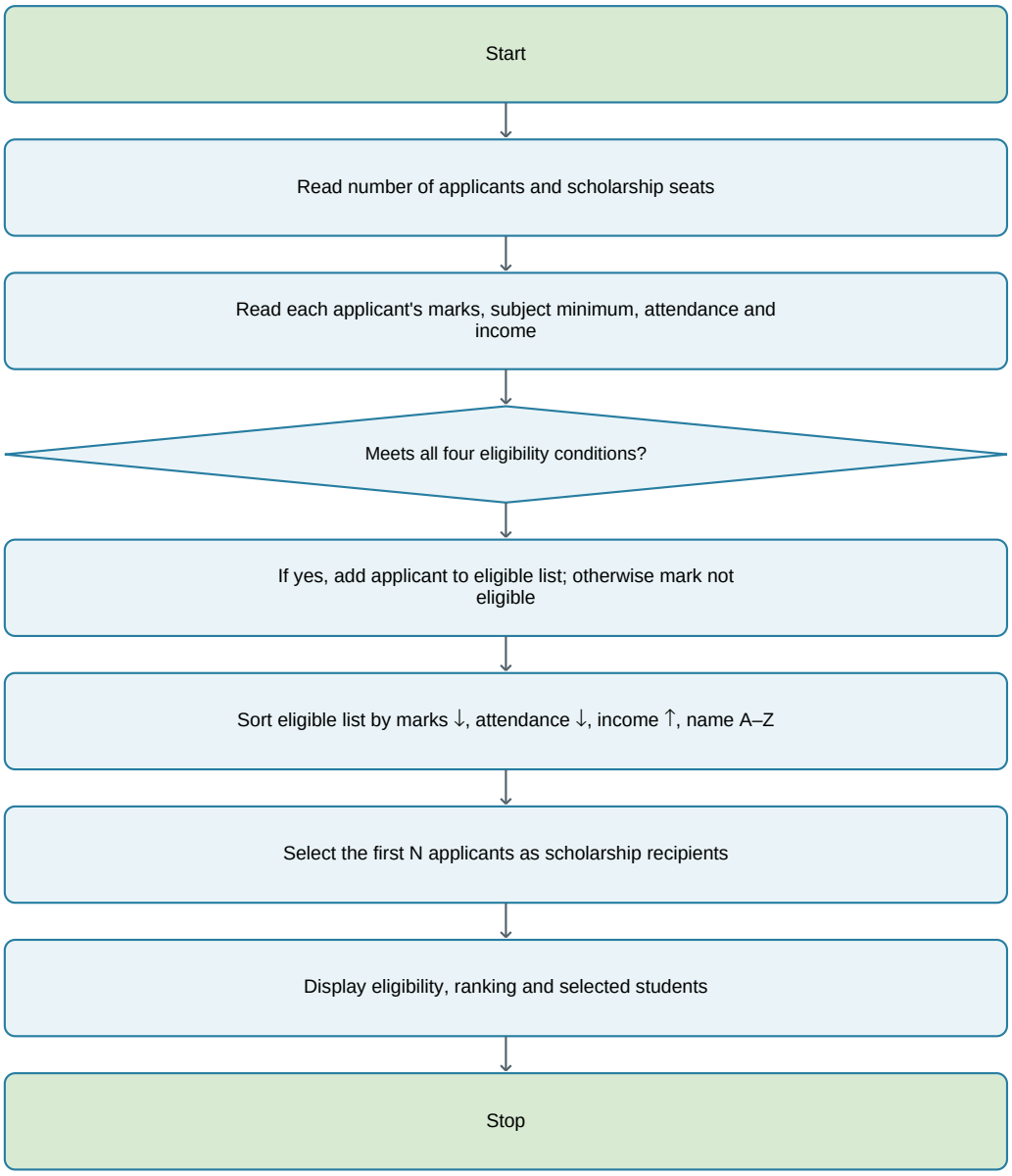
1. University Scholarship Allocation

Aim: Select eligible applicants using academic performance, subject minimums, attendance, and family income; rank them and apply tie-break rules.

Inputs: For each applicant: name, aggregate percentage, minimum subject score, attendance percentage, annual family income. Also enter scholarship seat count.

Rules / assumptions: Eligibility: aggregate ≥ 75, every subject ≥ 60, attendance ≥ 80%, and annual income ≤ ■2,50,000. Rank eligible students by aggregate (descending), then attendance (descending), then income (ascending), then name (A–Z).

Flowchart



Pseudocode

```
START
INPUT applicant_count, scholarship_seats
eligible_list = empty
FOR each applicant
    INPUT name, aggregate, subject_min, attendance, income
    IF aggregate >= 75 AND subject_min >= 60
       AND attendance >= 80 AND income <= 250000 THEN
        ADD applicant to eligible_list
    ELSE
        MARK applicant as NOT ELIGIBLE
    ENDIF
ENDFOR
```

```
SORT eligible_list BY aggregate DESC,
     attendance DESC, income ASC, name ASC
ASSIGN ranks starting at 1
SELECT first scholarship_seats applicants
DISPLAY eligibility, ranks and selected list
STOP
```

Sample Input

```
Applicants (name, aggregate %, lowest subject %, attendance %, income ■):
Asha, 88, 65, 92, 180000
Bala, 88, 62, 90, 200000
Charan, 72, 70, 95, 150000
Divya, 81, 58, 88, 120000
Seats = 2
```

Sample Output

```
Asha: Eligible, Rank 1, Selected
Bala: Eligible, Rank 2, Selected
Charan: Not eligible (aggregate below 75%)
Divya: Not eligible (subject minimum below 60%)
Scholarship recipients: Asha and Bala
```

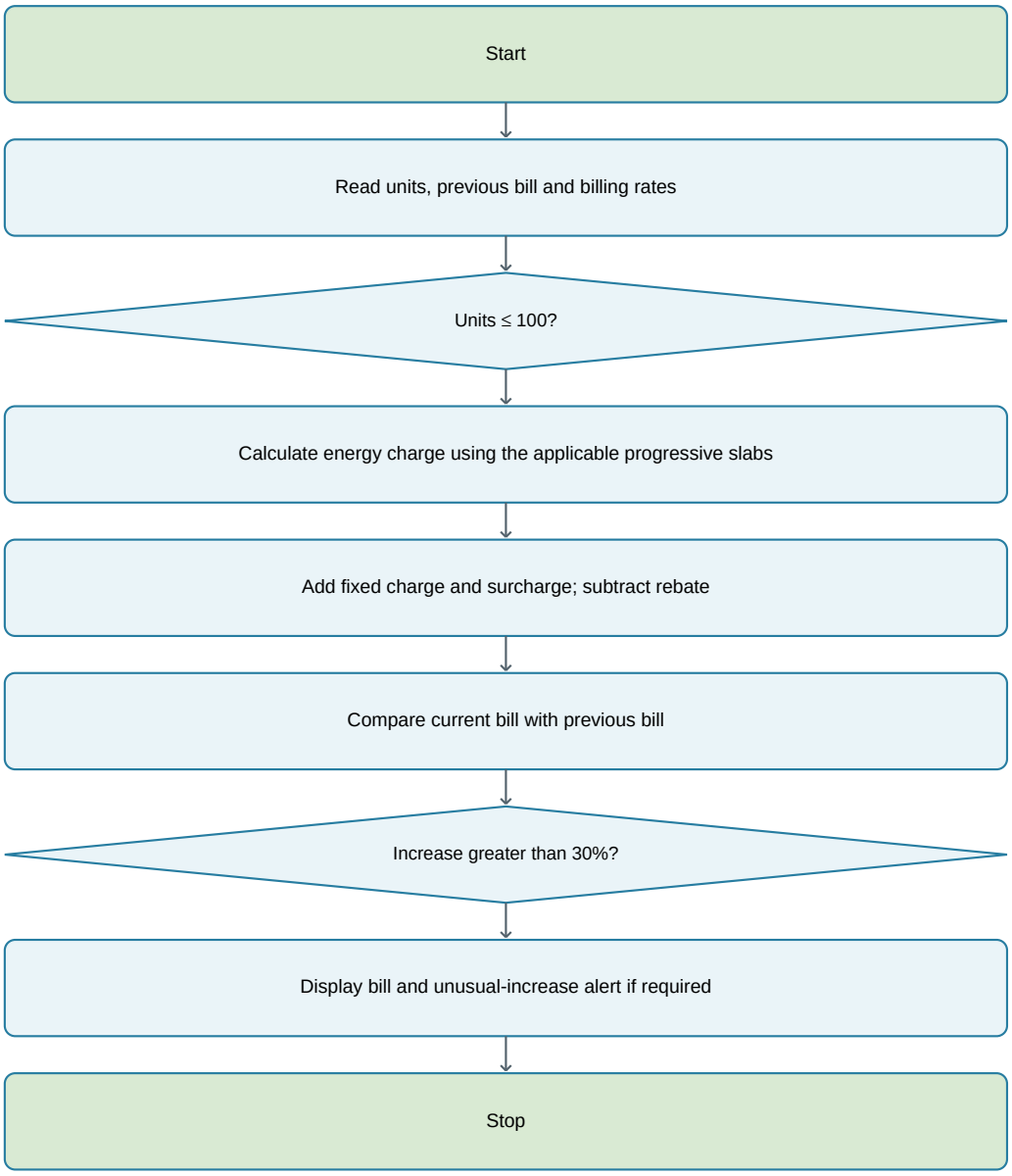
2. Progressive Electricity Billing

Aim: Calculate monthly electricity bills using consumption slabs, fixed charge, surcharge and rebate, then flag unusually large increases.

Inputs: Monthly units consumed, fixed charge, surcharge rate, rebate rate, and previous-month bill.

Rules / assumptions: Illustrative slabs: first 100 units at ■2/unit; next 100 at ■3/unit; units above 200 at ■5/unit. Add ■50 fixed charge. Add 5% surcharge to energy charge; apply 2% rebate to the subtotal. Flag a rise greater than 30% over the previous bill.

Flowchart



Pseudocode

```
START
INPUT units, previous_bill
IF units <= 100 THEN
    energy = units * 2
ELSE IF units <= 200 THEN
    energy = 100*2 + (units-100)*3
ELSE
    energy = 100*2 + 100*3 + (units-200)*5
ENDIF
surcharge = energy * 0.05
subtotal = energy + 50 + surcharge
rebate = subtotal * 0.02
```

```
bill = subtotal - rebate
IF previous_bill > 0 THEN
    increase = (bill-previous_bill)/previous_bill * 100
    IF increase > 30 THEN
        DISPLAY "Unusual increase"
    ENDIF
ENDIF
DISPLAY energy, surcharge, rebate, bill
STOP
```

Sample Input

```
Units consumed = 250
Previous month bill = ■500
Fixed charge = ■50; surcharge = 5%; rebate = 2%
```

Sample Output

```
Energy charge = 100×2 + 100×3 + 50×5 = ■750
Surcharge = ■37.50; subtotal = ■837.50
Rebate = ■16.75
Current bill = ■820.75
Increase over previous bill = 64.15%
Alert: Unusual increase (more than 30%)
```

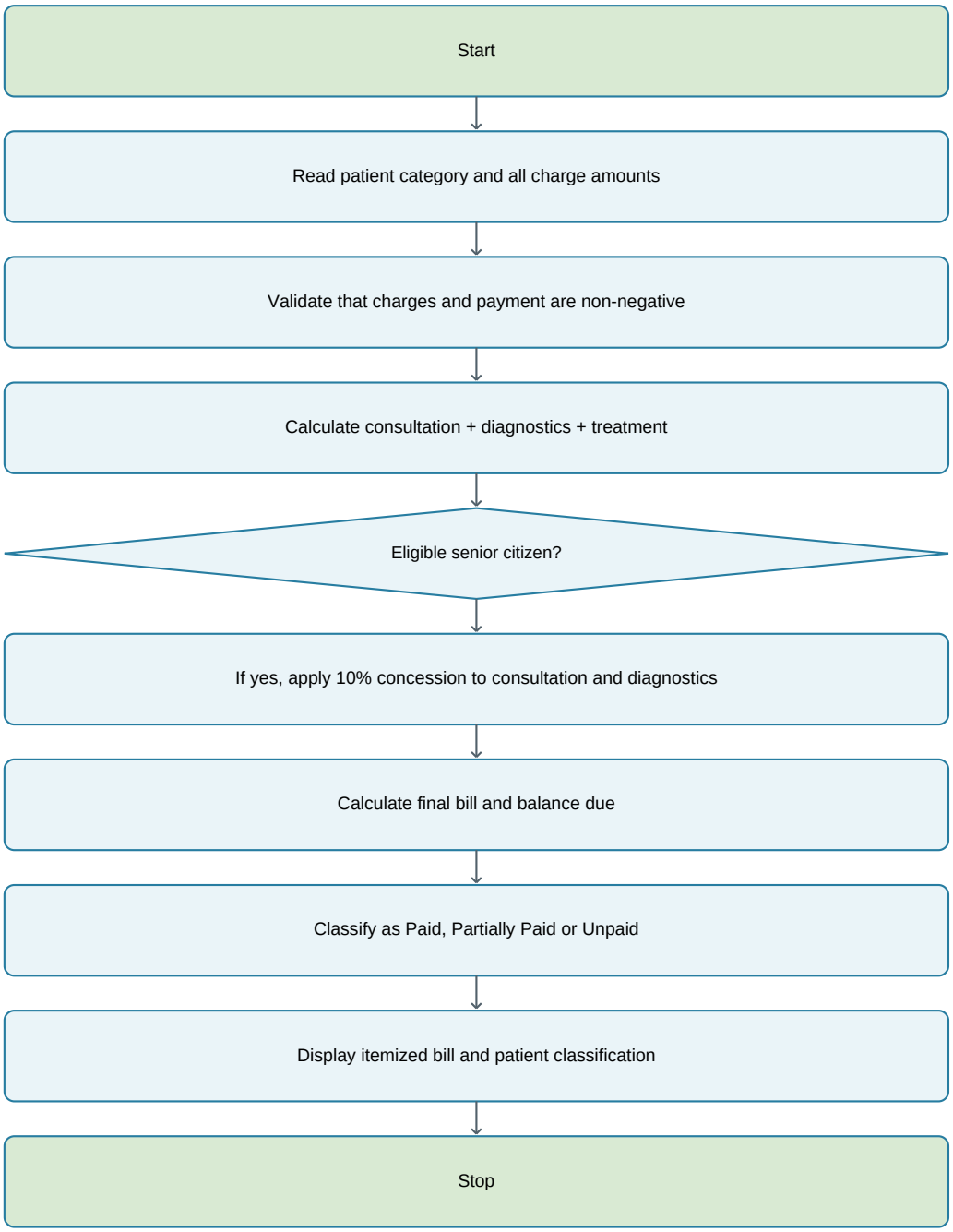
3. Hospital Billing and Patient Classification

Aim: Prepare a patient bill from consultation, diagnostic and treatment charges, apply a concession, validate the amount, and classify payment status.

Inputs: Patient type, consultation fee, diagnostic charges, treatment charges, concession eligibility, amount paid and payment policy.

Rules / assumptions: Illustrative policy: eligible senior citizen gets 10% concession on consultation and diagnostics only. Total cannot be negative. Classify as Paid if paid ≥ bill, Partially Paid if paid > 0 but less than bill, otherwise Unpaid.

Flowchart



Pseudocode

```
START
INPUT patient_type, consultation, diagnostics, treatment, paid
IF any charge < 0 OR paid < 0 THEN
    DISPLAY "Invalid input"; STOP
ENDIF
gross = consultation + diagnostics + treatment
```

```
IF patient_type = "Senior" AND concession_eligible THEN
    discount = (consultation + diagnostics) * 0.10
ELSE
    discount = 0
ENDIF
bill = gross - discount
balance = bill - paid
IF paid >= bill THEN
    status = "PAID"
ELSE IF paid > 0 THEN
    status = "PARTIALLY PAID"
ELSE
    status = "UNPAID"
ENDIF
DISPLAY gross, discount, bill, balance, status
STOP
```

Sample Input

```
Patient type = Senior (concession eligible)
Consultation = ■500; Diagnostics = ■1,000; Treatment = ■3,000
Amount paid = ■2,000
```

Sample Output

```
Gross charges = ■4,500
Concession = 10% of (■500 + ■1,000) = ■150
Final bill = ■4,350
Amount paid = ■2,000; balance due = ■2,350
Patient payment classification = PARTIALLY PAID
```

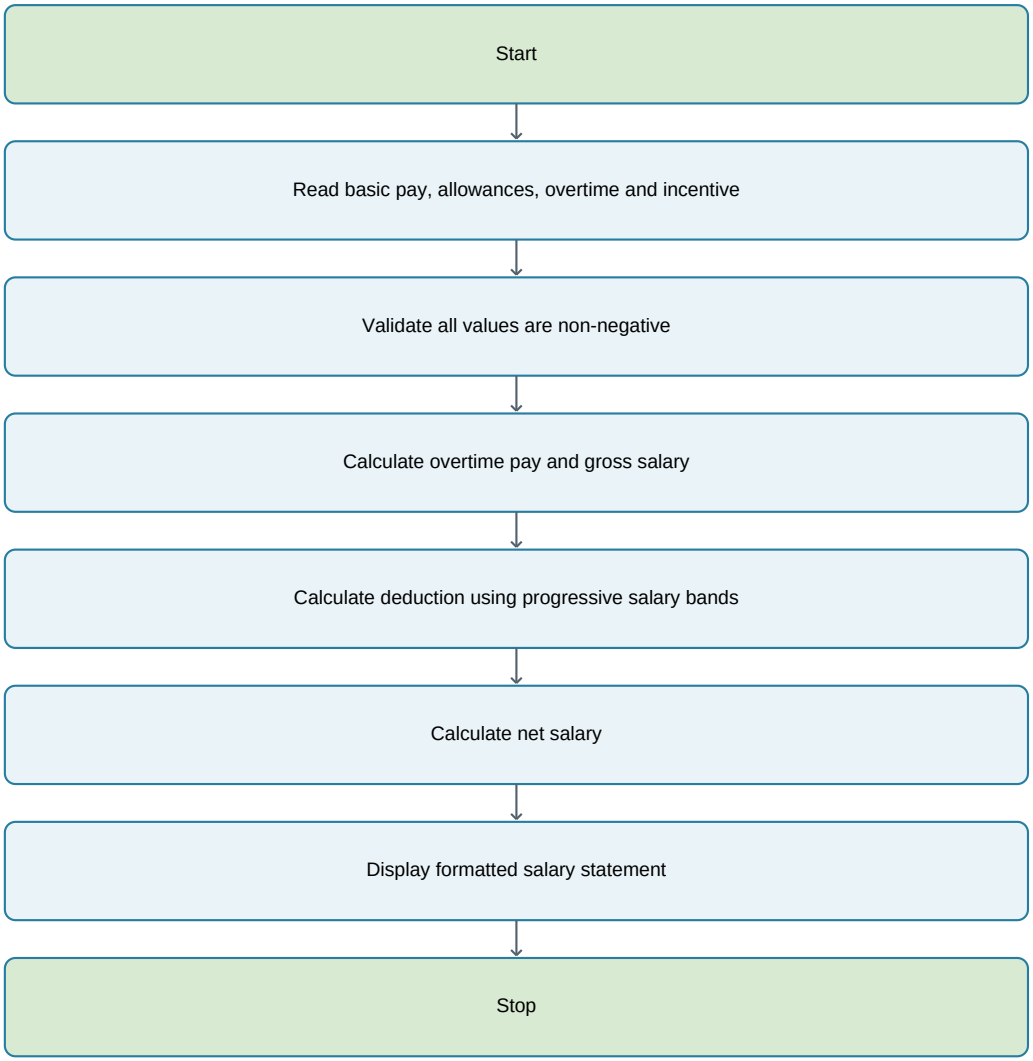
4. Employee Payroll and Deduction Analysis

Aim: Calculate gross salary from pay components, apply progressive deductions, and produce a salary statement.

Inputs: Basic pay, house rent allowance, other allowance, overtime hours, overtime rate and incentive.

Rules / assumptions: Gross = basic + allowances + overtime + incentive. Illustrative progressive deduction on gross: 0% up to ■15,000; 5% on the next ■10,000; 10% on the amount above ■25,000. Net salary = gross – deduction.

Flowchart



Pseudocode

```
START
INPUT basic, hra, other_allowance
INPUT overtime_hours, overtime_rate, incentive
IF any input < 0 THEN
    DISPLAY "Invalid input"; STOP
ENDIF
overtime_pay = overtime_hours * overtime_rate
gross = basic + hra + other_allowance
        + overtime_pay + incentive
IF gross <= 15000 THEN
    deduction = 0
ELSE IF gross <= 25000 THEN
    deduction = (gross-15000) * 0.05
ELSE
    deduction = 10000*0.05 + (gross-25000)*0.10
ENDIF
net = gross - deduction
DISPLAY gross, overtime_pay, deduction, net
STOP
```

Sample Input

```
Basic pay = ■20,000; HRA = ■5,000; Other allowance = ■2,000
Overtime = 10 hours at ■200/hour; Incentive = ■1,000
```

Sample Output

```
Overtime pay = ■2,000
Gross salary = ■30,000
Deduction = (■10,000×5%) + (■5,000×10%) = ■1,000
Net salary = ■29,000
```

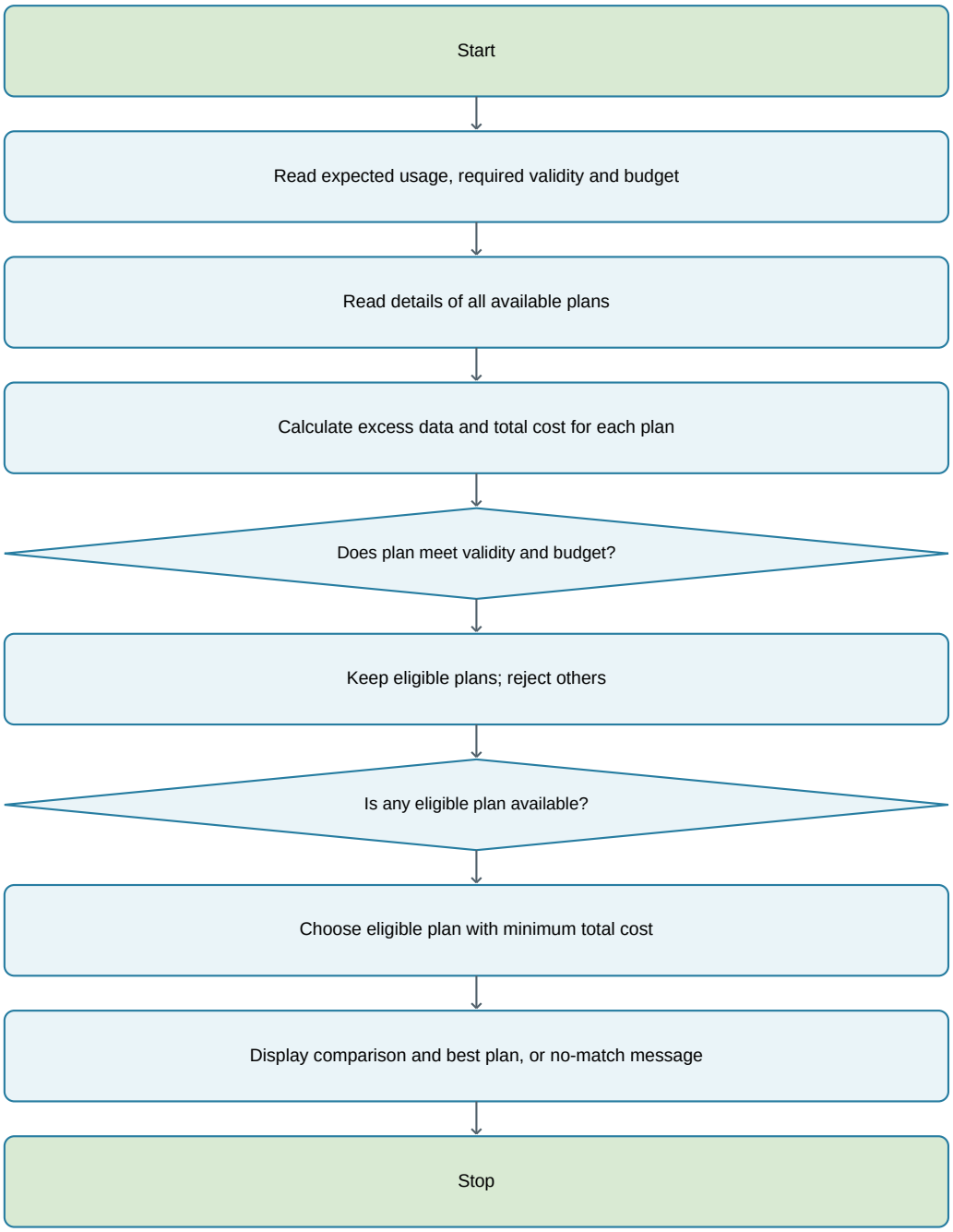
5. Multi-Plan Telecommunications Cost Optimization

Aim: Compare mobile plans against usage, validity and budget requirements, include excess-use charges, and select the lowest-cost eligible plan.

Inputs: Expected data usage, required validity in days, maximum budget, and each plan's price, data allowance, validity and excess-data rate.

Rules / assumptions: A plan is eligible when its validity meets the required days and its total cost is within budget. Total cost = plan price + excess data × excess rate. Select the eligible plan with the lowest total cost.

Flowchart



Pseudocode

```
START
INPUT usage_GB, required_days, budget
INPUT plan_count
best_plan = NONE
best_cost = infinity
FOR each plan
```

```
    INPUT name, price, data_GB, validity_days, excess_rate
    excess = MAX(0, usage_GB - data_GB)
    total = price + excess * excess_rate
    IF validity_days >= required_days AND total <= budget THEN
        IF total < best_cost THEN
            best_cost = total
            best_plan = name
        ENDIF
    ENDIF
ENDFOR
IF best_plan = NONE THEN
    DISPLAY "No plan meets the requirements"
ELSE
    DISPLAY best_plan, best_cost
ENDIF
STOP
```

Sample Input

```
Required usage = 5 GB; validity = 28 days; budget = ■300
Plan A: ■199, 3 GB, 28 days, excess ■20/GB
Plan B: ■249, 6 GB, 28 days, excess ■20/GB
Plan C: ■179, 2 GB, 14 days, excess ■15/GB
```

Sample Output

```
Plan A: ■199 + (2×■20) = ■239; eligible
Plan B: ■249 + ■0 = ■249; eligible
Plan C: ■179 + (3×■15) = ■224; rejected (validity only 14 days)
Most economical eligible plan: Plan A at ■239
```

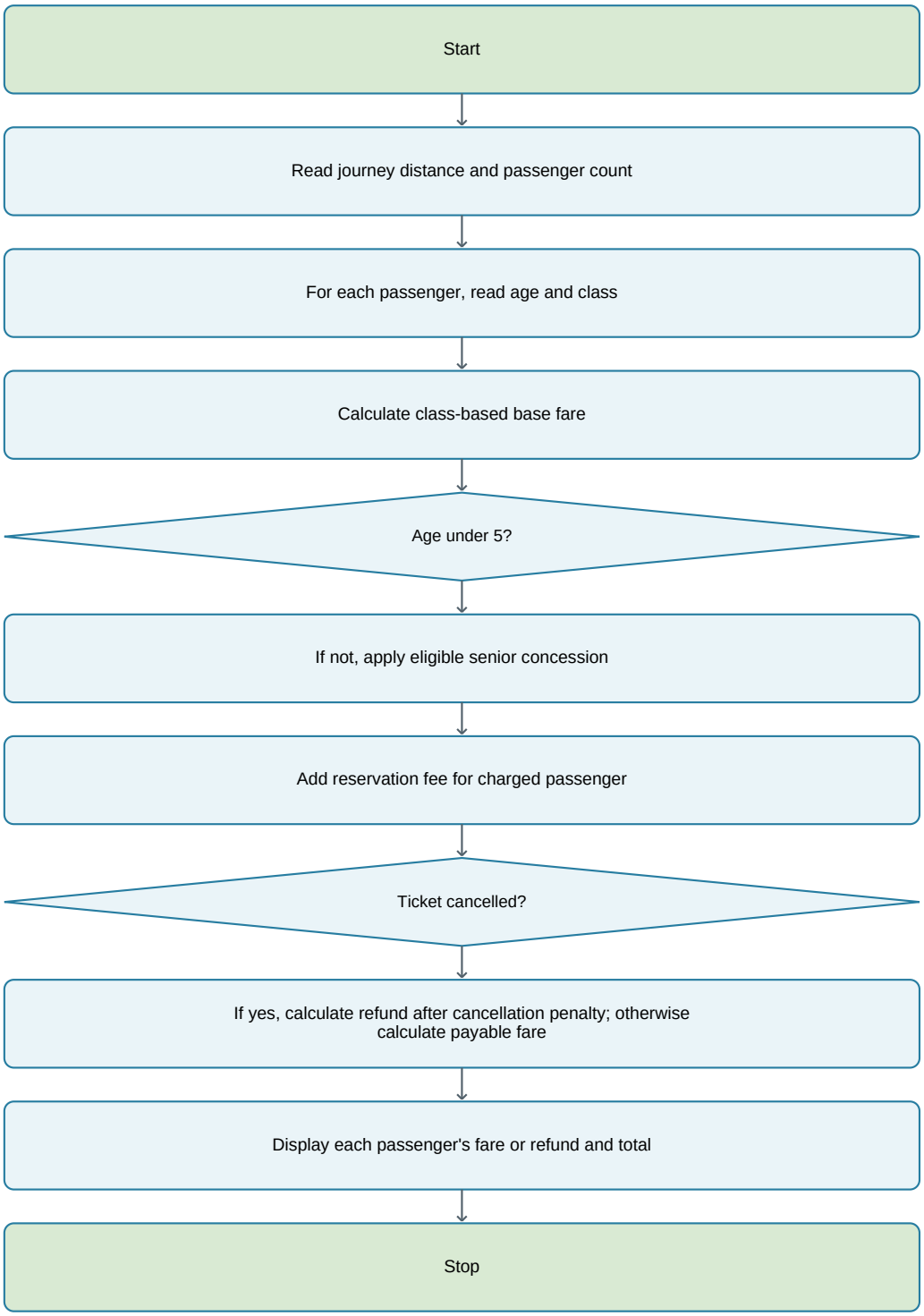
6. Railway Fare and Reservation Calculator

Aim: Calculate fares for several passengers using distance slabs, travel class, age concessions, reservation fees and cancellation penalties.

Inputs: Distance, class, passenger age, base fare rate per km, reservation fee, concession percentage and cancellation status.

Rules / assumptions: Illustrative rates: Sleeper ■1.00/km, AC ■2.00/km. Base fare = distance × class rate. Children under 5 travel free; passengers aged 60+ receive 20% base-fare concession. Add ■40 reservation fee per charged passenger. If cancelled, deduct 25% of base fare as cancellation penalty from the refundable base fare; reservation fee is non-refundable.

Flowchart



Pseudocode

```
START
INPUT distance, passenger_count
total_payable = 0
FOR each passenger
    INPUT age, travel_class, cancelled
    IF travel_class = "Sleeper" THEN rate = 1
    ELSE IF travel_class = "AC" THEN rate = 2
    ELSE DISPLAY "Invalid class"; CONTINUE
    ENDIF
    base = distance * rate
    IF age < 5 THEN
        base = 0
        reservation = 0
    ELSE
        IF age >= 60 THEN base = base * 0.80
        reservation = 40
    ENDIF
    IF cancelled THEN
        penalty = base * 0.25
        refund = base - penalty
        DISPLAY refund, "Reservation fee not refundable"
    ELSE
        fare = base + reservation
        total_payable = total_payable + fare
        DISPLAY fare
    ENDIF
ENDFOR
DISPLAY total_payable
STOP
```

Sample Input

```
Distance = 200 km
Passenger 1: age 30, Sleeper, not cancelled
Passenger 2: age 65, AC, not cancelled
Passenger 3: age 4, Sleeper, not cancelled
```

Sample Output

```
Passenger 1: Base ■200 + reservation ■40 = ■240
Passenger 2: AC base ■400; 20% concession = ■80; reservation ■40; fare = ■360
Passenger 3: Under 5; fare = ■0
Total payable = ■600
```

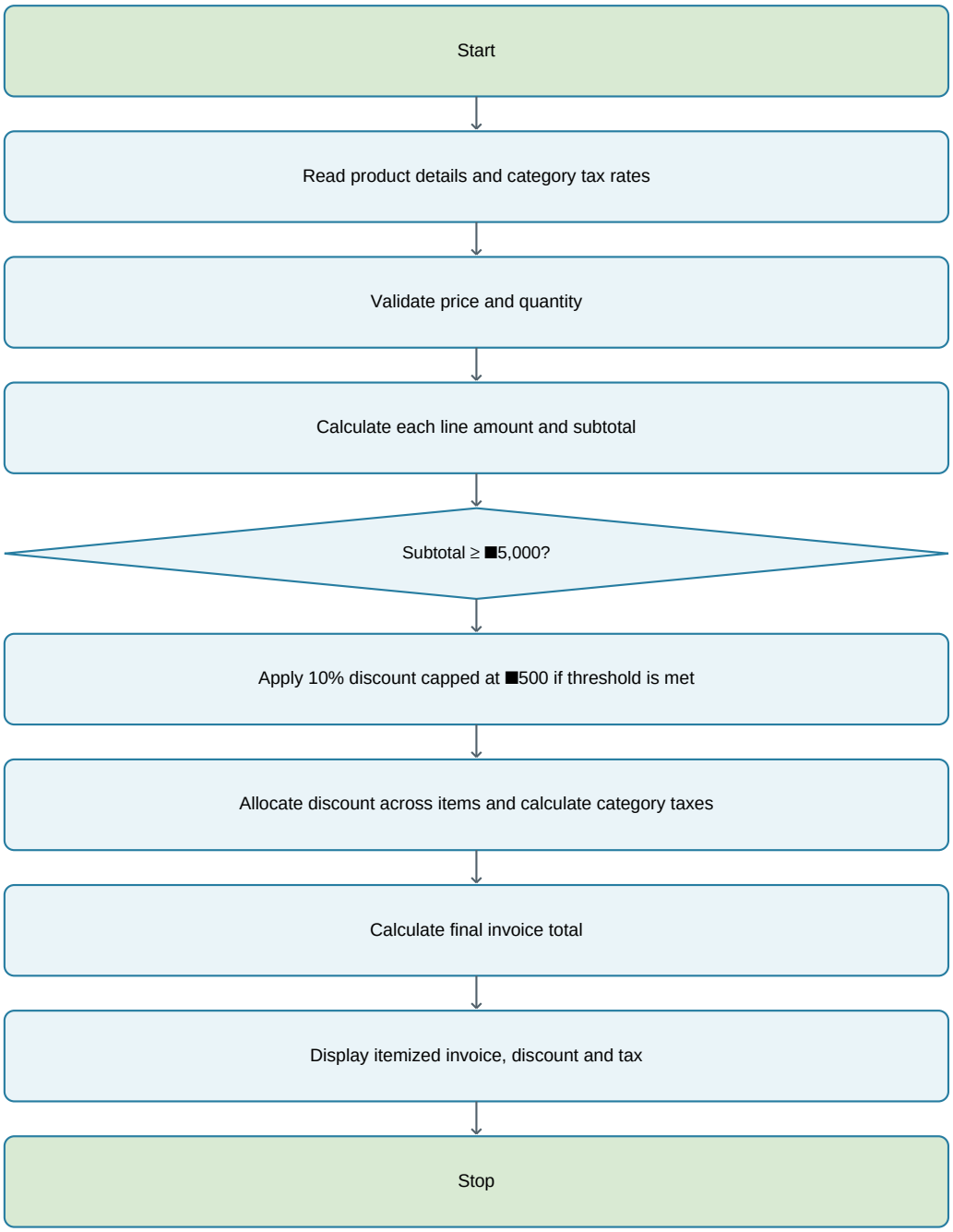
7. Retail Billing with Multiple Discount Rules

Aim: Generate an itemized retail invoice using quantities, category-specific taxes, threshold discounts and a maximum concession limit.

Inputs: For each item: name, category, unit price and quantity. Also enter discount threshold, discount rate, maximum discount and tax rate for each category.

Rules / assumptions: Line amount = price × quantity. Subtotal is the sum of line amounts. If subtotal ≥ ■5,000, discount is 10% of subtotal, capped at ■500. Apply category tax after discount, proportionally across items for this example. Final total = discounted subtotal + tax.

Flowchart



Pseudocode

```
START
INPUT item_count
subtotal = 0
FOR each item
    INPUT name, category, price, quantity, tax_rate
```

```
    IF price < 0 OR quantity < 0 THEN
        DISPLAY "Invalid item"; STOP
    ENDIF
    line = price * quantity
    subtotal = subtotal + line
    STORE item details
ENDFOR
IF subtotal >= 5000 THEN
    discount = MIN(subtotal * 0.10, 500)
ELSE
    discount = 0
ENDIF
discounted_subtotal = subtotal - discount
tax = 0
FOR each item
    share = item_line / subtotal
    discounted_line = item_line - discount * share
    tax = tax + discounted_line * item_tax_rate
ENDFOR
total = discounted_subtotal + tax
DISPLAY item lines, subtotal, discount, tax, total
STOP
```

Sample Input

```
Item 1: Notebook, Stationery, ■100, quantity 20, tax 5%
Item 2: Bag, Accessories, ■1,500, quantity 3, tax 12%
Discount rule: 10% if subtotal ≥ ■5,000; maximum discount ■500
```

Sample Output

```
Notebook line = ■2,000; Bag line = ■4,500
Subtotal = ■6,500
Discount = min(10% of ■6,500, ■500) = ■500
Discounted subtotal = ■6,000
Allocated discounted values: Notebook ■1,846.15; Bag ■4,153.85
Tax ≈ ■92.31 + ■498.46 = ■590.77
Invoice total ≈ ■6,590.77
```

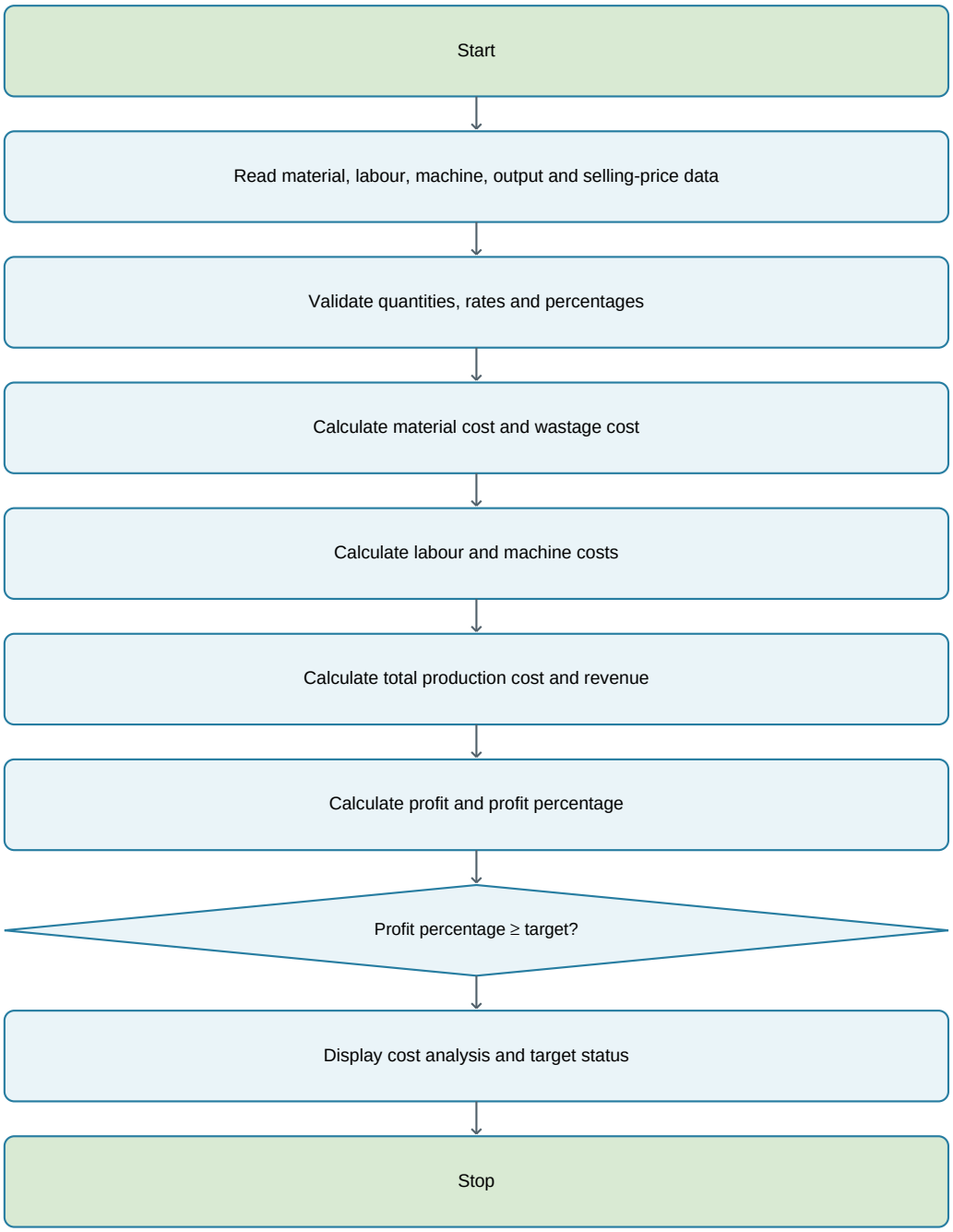
8. Manufacturing Cost and Profitability Analysis

Aim: Calculate total production cost, unit cost, selling revenue and profit; determine whether the required profit target is achieved.

Inputs: Material quantity and rate, labour hours and hourly rate, wastage percentage, machine cost, production quantity, selling price per unit and target profit percentage.

Rules / assumptions: Material cost = quantity × rate. Wastage cost = material cost × wastage %. Total cost = material + wastage + labour + machine. Profit = revenue – total cost. Profit percentage = profit ÷ total cost × 100.

Flowchart



Pseudocode

```
START
INPUT material_qty, material_rate, wastage_pct
INPUT labour_hours, labour_rate, machine_cost
INPUT units_produced, selling_price, target_pct
IF invalid negative input OR units_produced <= 0 THEN
    DISPLAY "Invalid input"; STOP
```

```
ENDIF
material = material_qty * material_rate
wastage = material * wastage_pct / 100
labour = labour_hours * labour_rate
total_cost = material + wastage + labour + machine_cost
revenue = units_produced * selling_price
profit = revenue - total_cost
profit_pct = profit / total_cost * 100
IF profit_pct >= target_pct THEN
    status = "TARGET ACHIEVED"
ELSE
    status = "TARGET NOT ACHIEVED"
ENDIF
DISPLAY total_cost, revenue, profit, profit_pct, status
STOP
```

Sample Input

```
Material: 100 kg at ■50/kg; wastage = 5%
Labour: 20 hours at ■100/hour; machine cost = ■1,000
Production = 100 units; selling price = ■80/unit
Target profit = 20%
```

Sample Output

```
Material cost = ■5,000; wastage = ■250
Labour = ■2,000; machine = ■1,000
Total cost = ■8,250
Revenue = 100×■80 = ■8,000
Profit = –■250 (loss); profit percentage = –3.03%
Target status: NOT ACHIEVED
```

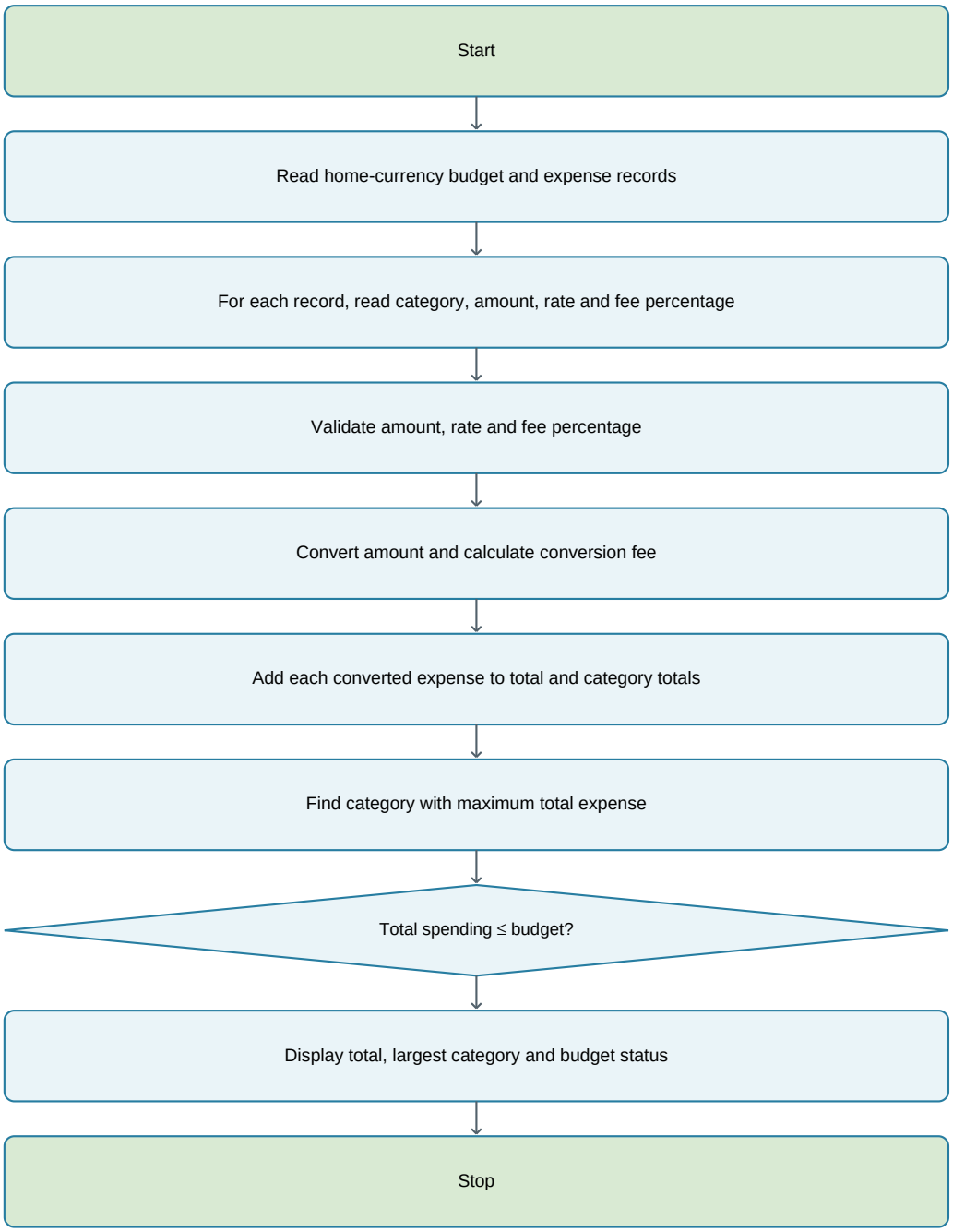
9. Multi-Currency Travel Budget Analysis

Aim: Convert travel expenses into the home currency using exchange rates and conversion fees, compare spending with budget, and find the largest expense category.

Inputs: Home-currency budget; for each expense: category, foreign-currency amount, exchange rate and conversion-fee percentage.

Rules / assumptions: Home-currency cost = foreign amount × exchange rate. Fee = converted cost × fee rate. Total category cost = converted cost + fee. Compare total with budget and identify the category with the highest home-currency cost.

Flowchart



Pseudocode

```
START
INPUT budget
INPUT expense_count
total = 0
largest_category = NONE
largest_cost = 0
```

```
FOR each expense
    INPUT category, foreign_amount, rate, fee_pct
    IF foreign_amount < 0 OR rate <= 0 OR fee_pct < 0 THEN
        DISPLAY "Invalid expense"; STOP
    ENDIF
    converted = foreign_amount * rate
    fee = converted * fee_pct / 100
    cost = converted + fee
    ADD cost to category_total[category]
    total = total + cost
ENDFOR
FOR each category
    IF category_total[category] > largest_cost THEN
        largest_cost = category_total[category]
        largest_category = category
    ENDIF
ENDFOR
IF total <= budget THEN status = "WITHIN BUDGET"
ELSE status = "OVER BUDGET"
ENDIF
DISPLAY total, largest_category, largest_cost, status
STOP
```

Sample Input

```
Budget = ■50,000
Hotel: USD 200, rate ■83/USD, fee 2%
Food: EUR 100, rate ■90/EUR, fee 2%
Transport: USD 100, rate ■83/USD, fee 2%
```

Sample Output

```
Hotel: ■16,600 + ■332 fee = ■16,932
Food: ■9,000 + ■180 fee = ■9,180
Transport: ■8,300 + ■166 fee = ■8,466
Total spending = ■34,578
Largest expense category = Hotel (■16,932)
Budget status = WITHIN BUDGET; remaining budget = ■15,422
```

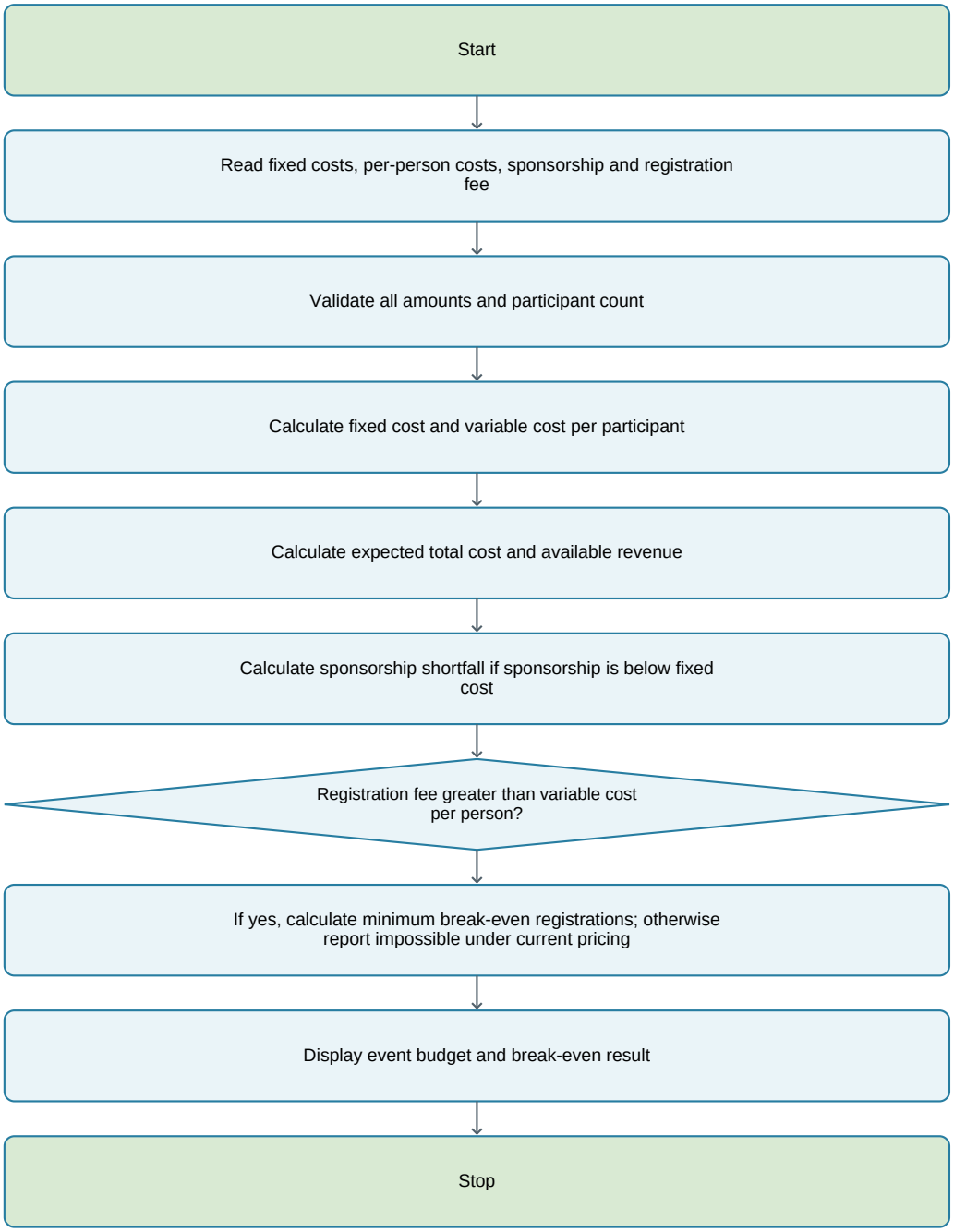
10. Campus Event Financial Planning

Aim: Estimate event expenses, participant contributions and sponsorship shortfall, then calculate the minimum registrations needed to break even.

Inputs: Fixed venue/equipment/transport costs, per-person catering/certificate costs, sponsorship amount, registration fee and expected participant count.

Rules / assumptions: Fixed cost = venue + equipment + transport. Variable cost per participant = catering + certificate. Total cost = fixed cost + variable cost × participants. Revenue = sponsorship + registration fee × participants. Break-even registrations = ceiling((fixed cost – sponsorship) ÷ (registration fee – variable cost per participant)), provided registration fee is greater than variable cost.

Flowchart



Pseudocode

```
START
INPUT venue, equipment, transport
INPUT catering_per_person, certificate_per_person
INPUT sponsorship, registration_fee, expected_people
```

```
fixed = venue + equipment + transport
variable = catering_per_person + certificate_per_person
total_cost = fixed + variable * expected_people
revenue = sponsorship + registration_fee * expected_people
shortfall = MAX(0, fixed - sponsorship)
contribution = registration_fee - variable
IF contribution <= 0 THEN
    break_even = "Not possible at this fee"
ELSE
    break_even = CEILING((fixed-sponsorship)/contribution)
ENDIF
IF revenue >= total_cost THEN status = "SURPLUS"
ELSE status = "DEFICIT"
ENDIF
DISPLAY total_cost, revenue, shortfall, break_even, status
STOP
```

Sample Input

```
Venue = ■10,000; equipment = ■3,000; transport = ■2,000
Catering = ■150/person; certificate = ■20/person
Sponsorship = ■5,000; registration fee = ■300/person
Expected participants = 50
```

Sample Output

```
Fixed cost = ■15,000; variable cost/person = ■170
Total cost = ■15,000 + (50×■170) = ■23,500
Revenue = ■5,000 + (50×■300) = ■20,000
Sponsorship shortfall against fixed costs = ■10,000
Contribution per registration = ■130
Break-even registrations = ceiling((■15,000−■5,000)/■130) = 77
At 50 registrations: DEFICIT of ■3,500
```